

ML Practical 2

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Section- Batch: B_B2

Roll No.: 22

```
In [8]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import
train_test_split from sklearn.linear_model import
LinearRegression from sklearn import metrics
```

```
In [10]: df=
pd.read_csv("USA_Housing.csv")
df.head()
```

Out[10]:

	Avg. Area Income	Avg. Area House Age	AreaAvg. Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price
20	0 79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06

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							V
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06	

2	61287.067179	5.865890	8.512727	5.13	36882.159400	1.058988e+06	g nDa W
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3	63345.240046	7.188236	5.586729	3.26	34310.242831	1.260617e+06	
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4	59982.197226	5.040555	7.839388	4.23	26354.109472	6.309435e+05	188 i 0
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```
In [12]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Avg. Area Income                      5000 non-null   float64
1   Avg. Area House Age                   5000 non-null   float64
2   Avg. Area Number of Rooms             5000 non-null   float64
3   Avg. Area Number of Bedrooms          5000 non-null   float64
```

```
4 Area Population          5000 non-null    float64
5 Price                    5000 non-null    float64
6 Address                  5000 non-null    objectdtypes: float64(6),
object(1) memory usage: 273.6+ KB
```

```
In [13]: df.isnull().sum()
```

```
Out[13]: Avg. Area Income          0
Avg. Area House Age              0
Avg. Area Number of Rooms        0
Avg. Area Number of Bedrooms     0
Area Population                  0
Price                            0
Address                          0
dtype: int64
```

```
In [14]: df.drop("Address",axis =1 , inplace=True)
```

```
In [15]: df.isnull().sum()
```

```
Out[15]: Avg. Area Income          0
Avg. Area House Age              0
Avg. Area Number of Rooms        0
Avg. Area Number of Bedrooms     0
Area Population                  0
Price                            0
dtype: int64
```

```
In [16]: df.describe()
```

```
Out[16]:
```

	Avg. AreaIncomeHouse Age	Avg. AreaNumber of	Avg. AreaRooms	Number ofBedrooms	Avg. AreaPopulationArea	
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5.000
mean	68583.108984	5.977222	6.987792	3.981330	36163.516039	1.232
std	10657.991214	0.991456	1.005833	1.234137	9925.650114	3.531
min	17796.631190	2.644304	3.236194	2.000000	172.610686	
25%	61480.562388	5.322283	6.299250	3.140000	29403.928702	
50%	68804.286404	5.970429	7.002902	4.050000	36199.406689	
75%	75783.338666	6.650808	7.665871	4.490000	42861.290769	
max	107701.748378	9.519088	10.759588	6.500000	69621.713378	1.593
						9.975

1.232
1.471
2.469

```
In [17]: pd.set_option('display.float_format', '{:.2f}'.format)

In [18]: df.head()
```

Out[18]:

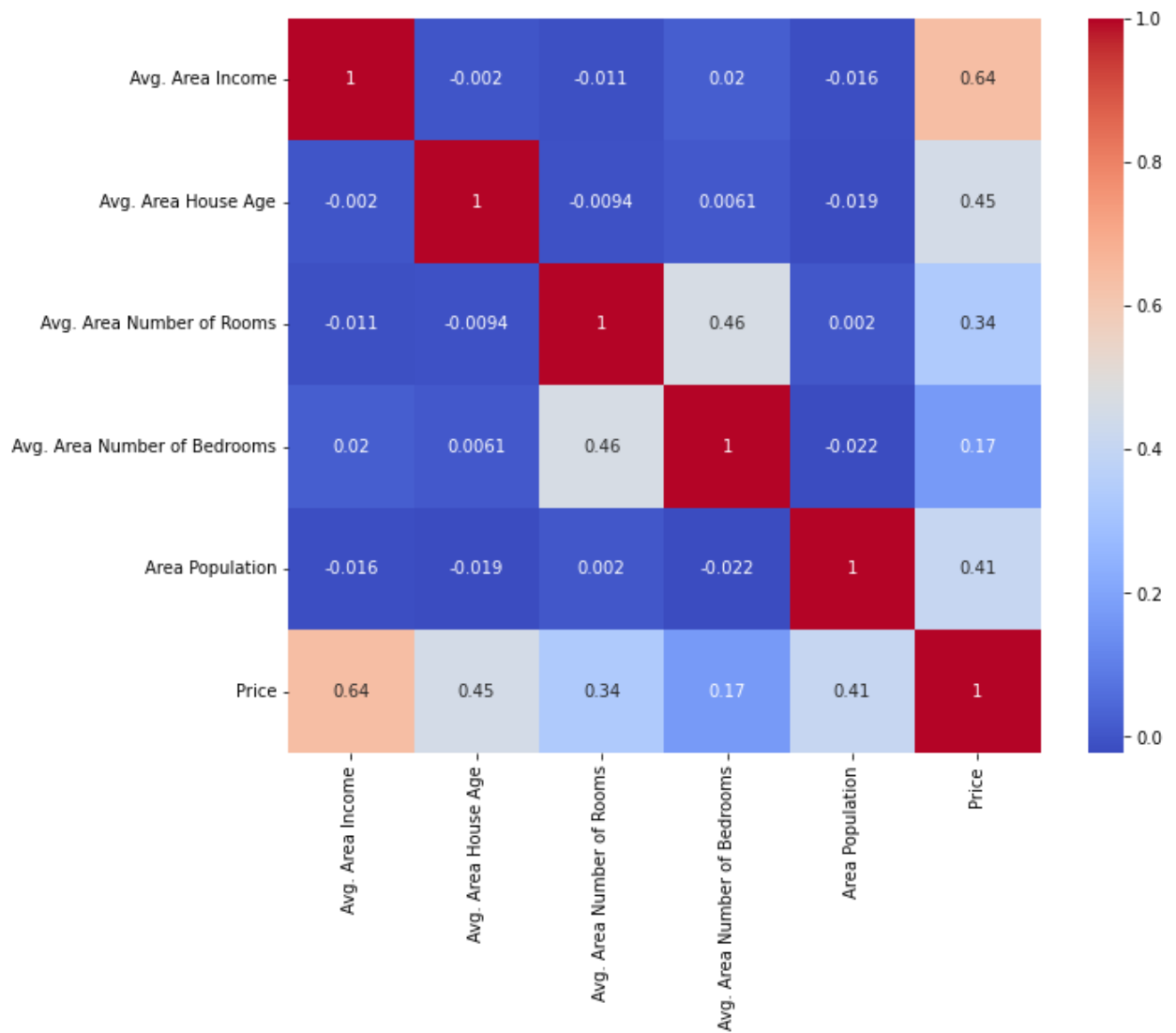
	IncomeAreaAvg.	HouseAreaAvg.	Number ofAvg. AreaRooms	Number ofBedroomsAvg. Area
--	----------------	---------------	-------------------------	----------------------------

0	79545.46	5.68	7.01	4.09	23086.80	1059033.56
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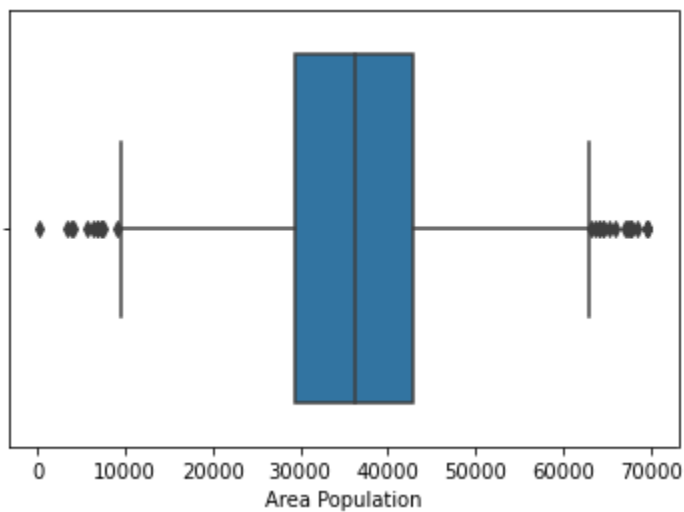
```
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(10, 8))
numeric_df = df.select_dtypes(include=['number'])
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.show()
```

1	79248.64	6.00	6.73	3.09	40173.07	1505890.91
2	61287.07	5.87	8.51	5.13	36882.16	1058987.99
3	63345.24	7.19	5.59	3.26	34310.24	1260616.81
4	59982.20	5.04	7.84	4.23	26354.11	630943.49

In [23]:

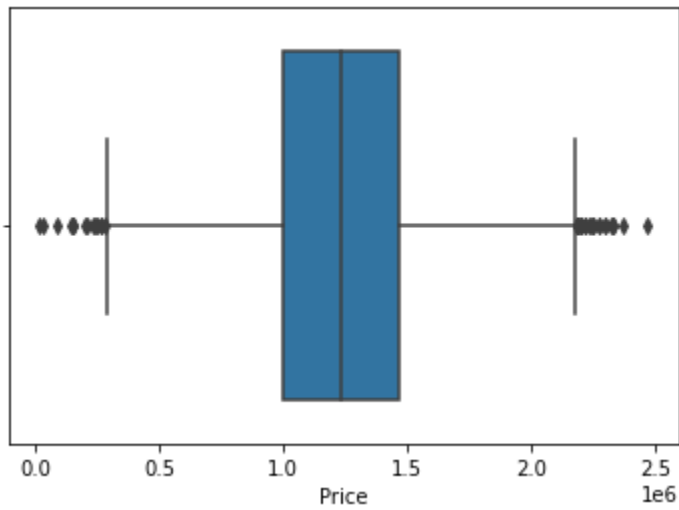


```
In [25]: sns.boxplot(x=df["Area Population"])
plt.show()
```



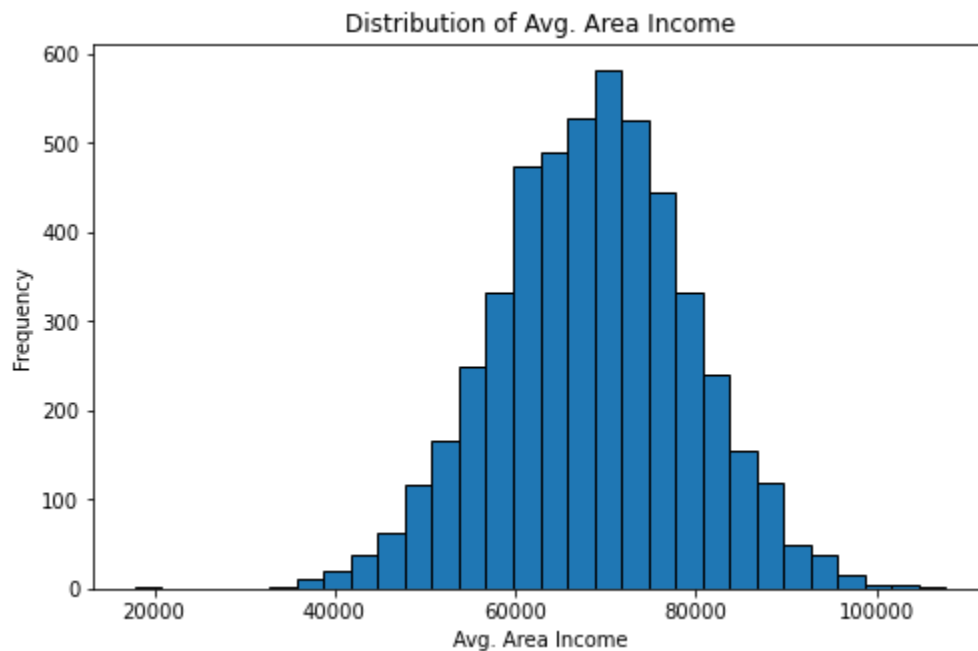
```
In [26]: sns.boxplot(x=df["Price"])
```

```
plt.show()
```



```
In [27]: import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(8,5))  
plt.hist(df['Avg. Area Income'], bins=30, edgecolor='black')  
plt.title('Distribution of Avg. Area Income')  
plt.xlabel('Avg. Area Income')  
plt.ylabel('Frequency')  
plt.show()
```



SLR

```
In [28]: X = df[['Avg. Area Income']]
         y = df['Price']
```

```
In [31]:
```

```
from sklearn.model_selection import train_test_split
X_train,X_test,          y_train,          y_test
In [32]: train_test_split(X,y,test_size=0.2,random_st
```

```
from sklearn.linear_model import LinearRegression
slr_model = LinearRegression()
slr_model.fit(X_train,y_train)
```

```
Out[32]:
```

```
y_pred =slr_model.predict(X_test)
```

```
LinearRegression() In [33]:
```

```
In [40]: from sklearn.metrics import mean_absolute_error
         from sklearn.metrics import mean_squared_error
         #from sklearn.metrics import
         root_mean_squared_error
         from sklearn.metrics import r2_score
         import numpy as np
```

```
print("Simple Linear Regression Results") print("-----
-----")
```

```
print("MAE :", mean_absolute_error(y_test, y_pred))
print("MSE :", mean_squared_error(y_test, y_pred))
print("RMSE :", np.sqrt(mean_squared_error(y_test,
y_pred))) print("R square score :",r2_score(y_test,
y_pred))
```

Simple Linear Regression Results

MAE : 217619.5644321996

MSE : 74402787880.48892

RMSE : 272768.74432472815

R square score : 0.41931484298724775

```
In [39]: print("Intercept:", slr_model.intercept_)
         print("Slope:",slr_model.coef_[0])
```

Intercept: -215047.7718841161

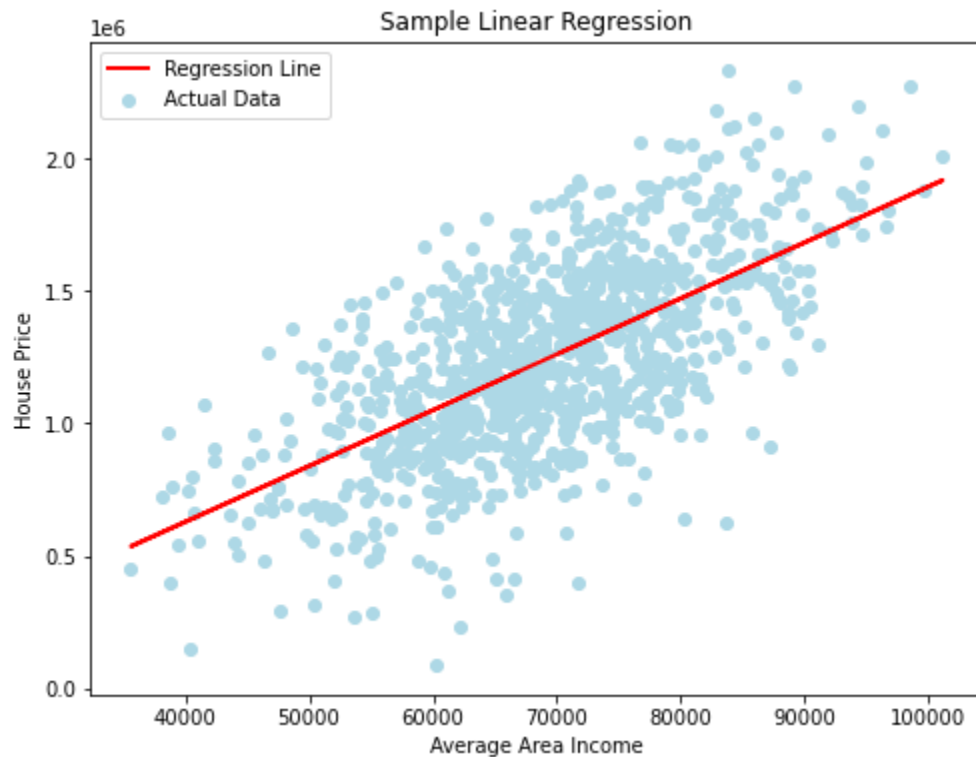
Slope: 21.088457310422807

```
In [60]: plt.figure(figsize=(8,6))

plt.scatter(X_test,y_test,color="lightblue",label = "Actual Data")

plt.plot(X_test,y_pred,color="red",linewidth=2,label ='Regression Line')

plt.xlabel("Average Area Income")
plt.ylabel("House Price")
plt.title("Sample Linear
Regression") plt.legend() plt.show()
```



```
In [61]: income = float(input("Enter Average Income:
")) new_data = np.array([[income]])
prediction = slr_model.predict(new_data)
print("Predict House Price =${:,.2f}".format(prediction[0]))
```

Enter Average Income: 53265.23
Predict House Price =\$908,233.76

In []: